

Presentation Script — AI in FinTech: Credit Scorecard System

Tools: MATLAB · Excel data · RiskLens dashboard

Before presenting: Open RiskLens (`cd fintech-dashboard → npm run dev`). Keep the report PDF as backup.

Printable PDF: <docs/presentation-script.pdf> — run `python3 scripts/generate_script_pdf.py` to refresh.

Slide map (follow the numbered sidebar)

Step	Page	Say this
1	Overview	Introduction & data
2	WOE & Binning	How we group clients by risk
3	Scorecard	Points that build the score
4	ROC & Validation	Is the scorecard reliable?
5	Portfolio Risk	Scores for 20 new applicants
6	Accept / Reject	Who gets a loan
7	Expected Loss	Money at risk in dollars
8	Risk Pricing	Minimum fair interest rate
9	Model Performance	Overall results
10	Final Decision	Final portfolio
11	Live Client Test	Quick live demo
12	MATLAB Charts	Extra charts <i>(optional)</i>

Speaker 1 — Introduction

[Title slide — “AI in FinTech: Credit Scorecard & Portfolio Optimization”]

Good morning / good afternoon. We present **Portfolio Optimisation and Pricing Interest Rate by Using a Scoring Model** — how a bank can use **AI in FinTech** to score loan applicants, decide who to lend to, and set a fair interest rate.

[Switch to Slide 1 — Overview (/)]

RiskLens is our dashboard. It shows the same results as our MATLAB work and written report — one place to see the full story.

[Point at the four cards at the top.]

In short:

- **500** past clients used to learn patterns
- **20** new applicants we score today
- **14 accepted, 6 rejected**
- About **\$129,330** total expected loss on **\$1.4 million** lent

[Point at the workflow strip — do not read every step.]

The idea is simple: learn from past clients in Excel, build a **credit score from 0 to 100** in MATLAB, then use RiskLens to show **accept or reject**, **expected loss**, and **minimum interest rate**. Higher score means safer client.

[Pause. Eye contact.]

Next we explain **why banks need this** and **what data we used**.

Handoff → Speaker 2

“Let’s look at the business problem and our data.”

Speaker 2 — Problem & data

[Stay on Slide 1 — Overview]

When someone applies for a loan, the bank asks: **Will they pay back?**

If the bank says yes to too many risky people, it loses money. If it says no to too many safe people, it loses customers. A **credit scorecard** ranks people from safer to riskier using facts we already know — age, income, home status, job status.

[Scroll to Default Distribution — pie chart.]

We have **500 past clients** in Excel (`HistoricalData`). Each row has age, income, housing, employment, and whether they **defaulted** (failed to pay) or not.

[Point at the pie chart numbers.]

- **320** paid as agreed (good)
- **180** defaulted (bad)
- About **36%** defaulted overall — enough risk to justify a scoring tool

[Point at default rate by income.]

Lower income → **higher default rate**. Under **\$30,000**, about **57%** defaulted. Above **\$50,000**, about **21%**. So **income** matters a lot in our final model.

[Point at default rate by age.]

Younger clients default more often. Homeowners do better than renters. Employed clients do better than “Other” job status.

[Scroll to Portfolio applicants.]

We also have **20 new applicants** (`ActualPortfolioData`). Same fields, but we do not know yet if they will default — that is what we predict.

[State assumptions clearly.]

We use the same loan size for everyone:

- **Loan:** \$100,000
- **If default, bank recovers 60%** → **loss = 40%** of the loan (called **LGD**)

All dollar results use these rules.

[Optional: Slide 12 — MATLAB charts, default pie]

Same 320 vs 180 split, shown as a chart.

[Pause.]

Next: **how we turn this data into a score**.

Handoff → Speaker 3

“Here is how we built the scorecard.”

Speaker 3 — Building the scorecard

[Slide 1 — Overview: point briefly at the workflow strip.]

We used MATLAB’s **Credit Scorecard Toolbox**. The main steps: load Excel → group clients into risk buckets → turn that into **points** → get a score from **0 to 100**.

[Switch to Slide 2 — WOE & Binning (/woe)]

[Point at IV cards.]

MATLAB splits age and income into groups automatically. For each group we measure **Weight of Evidence (WOE)** — basically, **is this group safer or riskier than average?**

Factor	Strength
Income	Strong
Age	Medium
Employment	Medium
Housing	Weak

[Point at Income WOE chart.]

Low income → bars below zero → **higher risk**. High income → above zero → **lower risk**.

[Scroll to predictor tables.]

Each table shows the group, how many clients, default %, and WOE — same idea as in our report.

[Switch to Slide 3 — Scorecard (/scorecard)]

[Point at Income points chart.]

MATLAB adds **points** for each factor. Safer groups get more points. Example for income: best group about **+29 points**, worst about **−9 points**. Employed clients get more points than “Other”. Homeowners get more than renters.

Every applicant’s points add up to a **final score from 0 to 100**.

[Pause.]

The big question: **does the score actually separate good from bad clients?** Speaker 4 answers that.

Handoff → Speaker 4

“Next: checking the scorecard and scoring the 20 applicants.”

Speaker 4 — Validation & portfolio scoring

[Switch to Slide 4 — ROC & Validation (/validation)]

[Point at metric cards.]

We checked the scorecard on the same **500 historical clients**:

What it means	Value
AUC — how well the score ranks risky vs safe	0.6775
KS — how far good and bad clients separate	0.2892
Cutoff score — accept at or above this	48.20

[Point at ROC curve.]

The green curve is our model. The dashed line is random guessing. Our curve is **above** the line — the scorecard is **useful**, not perfect.

[Simple rule:]

Accept if score $\geq$ 48.20 · Reject if below

The cutoff comes from the data, not from a guess.

[Switch to Slide 5 — Portfolio Risk (/portfolio)]

[Point at scatter plot.]

Each dot is one of the **20 new applicants**. Left to right = score. Up = chance of default (**PD**).

[Point at red line at 48.20.]

Right of the line → **accepted**. Left → **rejected**.

[Green vs red dots.]

Higher score → lower default chance. That is what we want.

[Risk bands:]

- **Low risk** (score $\geq$ 80): **5** clients
- **Medium** (50–80): **9** clients
- **High** (below 50): **6** rejected

[Pause.]

Now we turn scores into **money and interest rates**.

Handoff → Speaker 5

“Here are the main results in dollars.”

Speaker 5 — Results

[Switch to Slide 6 — Accept / Reject (/decisions)]

[Point at breakdown chart.]

Rule: score ≥ 48.20

- 14 accepted
- 6 rejected
- 70% of applications would get a loan

[Point at bar chart sorted by score.]

Clients 4 and 5 score 100 (safest). Client 1 scores 0 (riskiest).

[Client 8 — borderline.]

Client 8 scores 55.84 — just above the line, so **accepted**, but still **medium risk** (below 80).

[Table — clients 4, 5, 1.]

Examples:

- Client 4: score 100, default chance 12.79%, accepted, expected loss \$5,114
- Client 1: score 0, default chance 70.59%, rejected

[Switch to Slide 7 — Expected Loss (/expected-loss)]

[Point at summary cards.]

For **accepted clients only**:

Total expected loss	\$129,329.63
Average default chance	23.09%
Total loans (exposure)	\$1,400,000

[Point at bar chart.]

Each bar is how much the bank might lose on that client. Client 8 has the highest bar among accepted clients (~\$13,402) because their default chance is highest among those still accepted.

[Switch to Slide 8 — Risk Pricing (/pricing)]

[Point at rate chart.]

Minimum fair rate formula:

Minimum rate = default chance $\times$ loss if default (40%)

[Point at average line.]

- Average minimum rate (accepted): **9.24%**
- Safest accepted (clients **4, 5**): **5.11%**
- Highest accepted (client **8**): **13.40%**
- Highest rejected (client **1**): **28.23%** — too high to lend profitably

[Optional: Slide 12 — MATLAB charts for loss and rates]

[Pause.]

Last: **what this means for the bank** and **wrap-up**.

Handoff → Speaker 6

“To close: meaning, limits, and conclusion.”

Speaker 6 — Discussion & conclusion

[Switch to Slide 9 — Model Performance (/performance)]

[Point at radar or summary.]

Overall: the scorecard **works well enough** to rank clients (AUC **0.6775**, KS **0.2892**). It is not perfect — real banks use many more data fields.

What this means for the bank:

1. **Better choices** — Rejecting 6 risky clients avoids very high default rates (up to **70%**) and large losses per loan.
2. **Fair pricing** — Safer clients can get rates near **5%**; riskier accepted clients near **13%**, not one price for everyone.
3. **Planning** — **\$129,330** expected loss helps the bank set reserves on the **\$1.4M** book.
4. **Clear decisions** — Every accept/reject ties back to score and facts on file.

[Switch to Slide 10 — Final Decision (/final)]

[Point at headline.]

Final portfolio: 14 clients, \$1.4M exposure, \$129,329.63 expected loss, average minimum rate 9.24%.

[Accepted vs rejected lists.]

5 low risk · 9 medium risk · 6 rejected (high risk)

[Switch to Slide 11 — Live Client Test (/test) — optional]

Pick **Client 4**, click **Run credit test**.

Dashboard shows: score **100**, default chance **12.79%**, accepted, loss **\$5,114**, rate **5.11%** — matches the report.

Limitations (say honestly):

- Only **four** factors — real banks use much more.
- Scores are based on **past** data; the economy can change.
- Same **\$100k** loan and **60%** recovery for everyone — real life is more detailed.

Still, we show a full FinTech flow: data → score → decision → loss → pricing, with **AI/tools** (MATLAB + dashboard), not a pure machine-learning black box.

[Slide 10 or title slide]

Conclusion — three points:

1. We built a **credit scorecard** on **500** past clients (MATLAB + Excel).
2. For **20** new applicants: **14 accepted**, cutoff **48.20**, **\$129,330** expected loss on **\$1.4M**.
3. Minimum rates on accepted loans from **5.11%** to **13.40%**, average **9.24%**.

Thank you. We welcome **questions**.

Q&A — short answers

Question	Answer
What is WOE?	A number that shows if a group is safer or riskier than average.
Why score 48.20?	It is the best split between good and bad clients on historical data (KS).
Why is income strongest?	Low income had the highest default rates in the data.
What is LGD?	40% — the part of the loan the bank loses if the client defaults.
Why is total loss \$129,329.63?	Sum of expected loss for all 14 accepted clients.
Where is the code?	matlab/FinTech_Scorecard_Project.m and the Final/ folder.

Before you present

- [] RiskLens open and working
- [] Report PDF open as backup
- [] Each speaker knows their slides (1–11)
- [] Screen text large enough to read from the back

Aligned with `Final/fintech (1).pdf` and RiskLens navbar steps 1–12.